ENRIQUE JOSE ACOSTA DE LA RUA



MECHATRONICS ENGINEER / MACHINES, MOTION AND LAB AUTOMATION

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PROFILE

Mechatronics engineer (MSc UPV, 2027) with an industrial-design engineering background. I build complete machines end to end (precision mechanics, motion control, electronics and software), fabricated in my own workshop. My BEng thesis was an automated liquid-dispensing machine for laboratories; my MSc thesis is a 3-axis macro-photogrammetry robot with custom firmware, a digital twin and a GPU reconstruction pipeline.

PROJECTS

Macro-photogrammetry robot — 2025–2027

MSc thesis, UPV. Self-initiated; designed, built and programmed alone

- 3-axis positioner for under €500: SFU1605 ball screw on MGN12 rails, capstan drive with a 3D-printed PETG strap on preloaded tapered bearings, GT2 turntable.
- Custom STM32F429 firmware: trapezoidal ramps, USB JSON protocol, watchdog and heartbeat safety, StallGuard sensorless homing (σ = 28 µm, commanded positions).
- AI-assisted software stack (~45k lines, ~570 tests): PTP camera control, GPU focus stacking, ArUco calibration, COLMAP + RoMa reconstruction. 36/36 views registered below 0.6 px in 4 of 5 scans.

Automated liquid-dispensing machine for labs — 2025

BEng thesis, UPCT

- Two self-designed, 3D-printed peristaltic pumps on NEMA 17 steppers; load cell + HX711 close the loop on dispensed mass; ESP32 with an OLED recipe interface.

Smart camera flash — 2026

Personal project. Prototype: hardware done, firmware integration in progress

- ESP32-S3 flash with Wi-Fi and BLE phone control and a remote-trigger camera dongle. 4-layer KiCad PCB made by JLCPCB, hand-soldered down to 0402 and QFN.

EXPERIENCE

Industrial design and technology engineering intern — 2024–2025

IDEA Design Mindset, Murcia

- In charge of the electronics (hand-built) and firmware of a 3D-printing filament recycling machine for a Kickstarter campaign.
- Team member on a lamp that won the iF Design Award, NY Product Design Awards and A' Design Award. CAD, functional prototypes, mechanical simulation, renders; dealt with Chinese suppliers and manufacturers.

Freelance engineering, fabrication and teaching — 2023–now

Biomedical, research and industrial clients

- Tens of bone photogrammetry models for a maker of custom anatomical models; 3D-printed insect traps for a Universidad de Murcia research group; a full Fusion 360 CAM and CNC course; SolidWorks and Fusion 360 classes.

EDUCATION

MSc Mechatronics Engineering — 2025–Mar 2027

Universitat Politècnica de València. Coursework: state-space control, FOC and SVPWM, MATLAB/Simulink, sensors, Omron PLC and SCADA, ABB and Siemens drives

BEng Industrial Design Engineering and Product Development — 2020–2025

Universidad Politécnica de Cartagena

Summer school: Embodied AI and Intelligent Systems — Jul 2026

Beihang University International Summer School, Hangzhou, China. Certificate of completion

SKILLS

Mechanical	Strongest area. SolidWorks, Fusion 360, Inventor, Rhino, Blender. Precision positioning (ball screws, MGN rails, capstan and belt drives, preloaded tapered bearings), kinematics, torque and critical-speed sizing, tolerances and fits, DFM, FEA.
Fabrication	Own workshop: FDM (multi-material, PET-CF, PETG, TPU, ABS), SLA, manual lathe, small CNC. Fusion 360 CAM, custom G-code generation, PCB and part sourcing.
Electronics	KiCad (4-layer), Altium. Hand soldering down to 0402 and QFN. ESP32 and ESP32-S3, STM32F429, Raspberry Pi; TMC2209 and StallGuard, steppers, HX711 and load cells, PWM LED drivers.
Control	State-space control, FOC, SVPWM, MATLAB, Simulink. Omron PLCs and SCADA (Sysmac Studio), ABB and Siemens drives, TI motor-control boards.
Software	Python (OpenCV, PyTorch, PySide6), embedded C++ (PlatformIO), JavaScript, C#, Git. I architect, review and test; AI coding agents write much of the code. Basic C on my own.
Imaging and 3D	COLMAP, RoMa, Ceres, TSDF fusion, ArUco calibration, gsplat (3DGS and 2DGS), focus stacking. RealityScan, Metashape. Macro optics, cross-polarised lighting.

LANGUAGES

Spanish	Native
English	C2, Cambridge certificate
French	Elementary; relearning
Italian	Basic

PRACTICAL

Spanish (EU) citizen, free to work in Switzerland and the EU (B permit through the employment contract in CH). MSc ends Feb–Mar 2027; available full-time from July 2027 and willing to relocate. Driving licence (B) since 2020.

BEYOND WORK

Building tech since my teens (first FPV racing drone at 16). Photography, hiking, exploring, travelling and mountain biking.